11/04/2026: Intro to Probability (Part 2)

CSCI 246: Discrete Structures

Textbook reference: Sec 31, Scheinerman

Announcements

- I'm giving the Applied Math Seminar tomorrow (Thursday, November 5th) at 3:10pm in Wilson Hall #1-144. Title: *Bayesian Multi-Sample Changepoint Inference with Linear Computational Cost.* Feel free to attend.
- I need to cancel office hours today. If you were planning on attending, send me an email so we can schedule an alternate time.

Today's Agenda

- Odds and Ends: Study guide, finish slide from Monday's deck ($\approx$ 5 mins)
- Mini-lecture ($\approx$ 25 mins)
- Group exercises ($\approx$ 20 mins)

Study Guide For Quiz #9

Group exercises. Know how to solve the group exercises from the course meetings on Functions (10/21), Big O and Big Theta (10/28), and Algorithmic Efficiency (10/30).

Readings.

Sec. 24 (Functions)

- Understand the defintion of a function, as in Examples 24.2 and 24.4.
- Know how to prove whether function is 1-to-1 – as in Examples 24.16 and 24.17, onto – as in Example 24.20, or both (that is a bijection) – as in Example 24.23.

Epp Sec 11.2 (Big-O, Big-Omega, Big-Theta Notations)

- Know how to show that a function is Big-O or Big-Omega of some other function by applying the definitions of Big-O and Big-Omega, as in Example 11.2.3.
- Understand the problem with Big O notation, as in Example 11.2.6.
- Know how to show that a function is Big-Theta of some other function by applying the definition of Big-Theta, as in Example 11.2.5a.
- Know how to apply the theorem on polynomial orders, as in Example 11.2.8.

Epp Sec 11.3 (Analysis of Algorithm Efficiency I)

- Know how to compute the order of an algorithmic segment via computing the number of elementary operations that are performed, as in Examples 11.3.2, 11.3.3, and 11.3.4.
- Understand the difference between worst-case, best-case, and average-case complexities, as presented in Example 11.3.1. (For more, understand Examples 11.3.6 and 11.3.7 at a high level.)

Finish slide from Monday's deck

Applications
Of The Equally Likely Probability Formula

Reminder of Proposition: Equally Likely Probability Formula

If (P, S) is a probability space in which all outcomes are equally likely and E is an event in S, then the probability of E is

$$P(E) = \frac{|E|}{|S|} = \frac{\text{the number of outcomes in } E}{\text{the number of outcomes in } S}$$

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$$S = \Big\{(1,1,1,1,1,1,1,1,1,1), (1,1,1,1,1,1,1,1,1,2), \ldots, (6,6,6,6,6,6,6,6,6,6)\Big\}$$

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Are all outcomes equally likely?

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$$P(E) = \frac{|E|}{|S|}$$

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$$P(E) = \frac{|E|}{|S|} = \frac{6}{6^{10}} = \left(\frac{1}{6}\right)^9 \approx \frac{1}{10 \text{ million.}}$$

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Now $|S| = 2^5$ and $|E| = \binom{5}{2}$.

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$$P(E) = \frac{|E|}{|S|} = \frac{\binom{5}{2}}{2^5} = \frac{10}{32} \approx 0.3125.$$

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A poker hand is called a *four of a kind* if four of the five cards show the same value (e.g. all 7s or all kings). What is the probability that a poker hand is a four of a kind?

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Let $E \subset S$ be the event that the poker hand is a four of a kind. Then by the equally likely probability formula,

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Now $|S| = \binom{52}{5}$ and $|E| = 13 \times 48$.

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Now $|S| = \binom{52}{5}$ and $|E| = 13 \times 48$. Hence,

$$P(E) = \frac{|E|}{|S|} = \frac{13 \times 48}{\binom{52}{5}} \approx 0.00024.$$

Final Answer: The probability of getting a four of a kind in a 5-card poker hand is approximately 0.00024, or about 1 in 4,167 hands.

Probability of Complements

The Birthday Problem

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A Few Basic Properties of Probability

Let (S, P) be a probability space. Then

(a) (Probability of entire space.) $P(S) = 1$.

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(c) (Probability of complement.) $P(\overline{A}) = 1 - P(A)$, where $\overline{A} := S - A$ is called the *complement* of A (i.e. A didn't happen).

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Solution. Let A be the event that at least two people in this room share a birthday. It's easier to work with the event $\overline{A}$ (that *everyone's birthday is different*). Suppose there are 25 people in this room. Then

$$P(A) = 1 - P(\overline{A}) = 1 - \frac{(365 \cdot 364 \cdots 341)}{365^{25}} \approx 0.57$$

Example: The Gambler's Puzzle (de Méré, 1654)

Which bet is more likely to win?

(a) At least one 6 in 4 rolls of a die.

(b) At least one double-six in 24 rolls of a pair of dice.

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Now $|S| = 6^4$ and $|\overline{E}| = 5^4$. Hence, by the complement property and the equally likely probability formula,

$$P(E) = 1 - P(\overline{E}) = 1 - \frac{|\overline{E}|}{|S|} = 1 - \left(\frac{5}{6}\right)^4 \approx 0.518.$$

Example: The Gambler's Puzzle (de Méré, 1654), continued

Bet (b). What is the sample space?

Example: The Gambler's Puzzle (de Méré, 1654), continued

Bet (b). Let S be the sample space. Each roll is a pair, so an outcome is a list of 24 pairs:

$$S = \{\big((1,1),\ldots,(1,1)\big),\ \big((1,1),\ldots,(1,2)\big),\ \ldots,\ \big((6,6),\ldots,(6,6)\big)\}$$

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Let $E \subset S$ be the event that at least one roll is $(6,6)$. Its complement is the set of lists in which every roll is one of the other 35 pairs:

$$\overline{E} = \{\big((1,1),\ldots,(1,1)\big),\ \big((1,1),\ldots,(1,2)\big),\ \ldots,\ \big((6,5),\ldots,(6,5)\big)\}$$

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Now $|S| = 36^{24}$ and $|\overline{E}| = 35^{24}$.

Example: The Gambler's Puzzle (de Méré, 1654), continued

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Now $|S| = 36^{24}$ and $|\overline{E}| = 35^{24}$. Hence, by the complement property and the equally likely probability formula,

$$P(E) = 1 - P(\overline{E}) = 1 - \frac{|\overline{E}|}{|S|} = 1 - \left(\frac{35}{36}\right)^{24} \approx 0.491.$$

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Answer. Bet (a) wins more often than it loses; bet (b) loses more often than it wins.

Warning:
Outcomes Aren't Always Equally Likely

Question

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- **The trap:** "There are 11 possible sums, so $\frac{1}{11}$."

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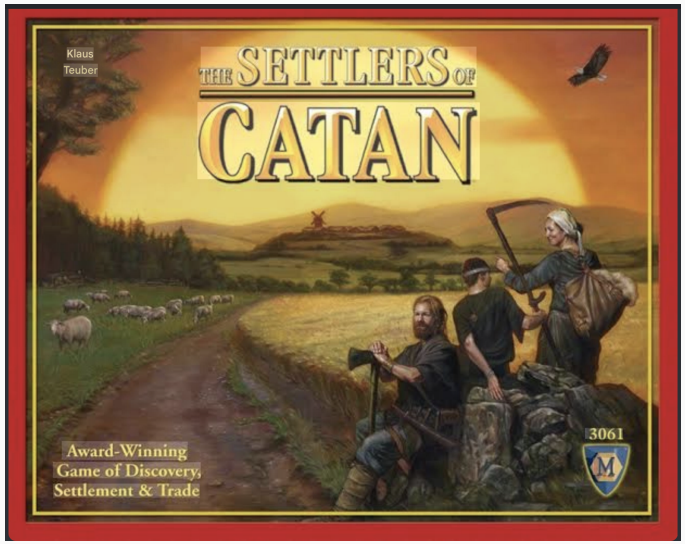
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- **Catan players know this:** the 6 and 8 tiles are the valuable ones.



Warning: Outcomes Aren't Always Equally Likely

Consider the probability space (S, P) where the sample space

$$S = \{A, B, C, D, F\}$$

is your possible grades in CSCI 246. Suppose that you have been performing well in the class so far, and the probability of you obtaining each grade is given by

$$P(A) = .50,\ P(B) = .30,\ P(C) = .10,\ P(D) = .07,\ P(F) = .03$$

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Here, the equally likely probability formula gives the wrong result:

$$P(E) \neq \frac{|E|}{|S|} = \frac{2}{5} = .40$$

General addition rule

Proposition: Basic Properties of Probability

Let (S, P) be a probability space. Then

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(c) (Probability of complement.) $P(\overline{A}) = 1 - P(A)$,
where $\overline{A} := S - A$ is called the *complement* of A (i.e. A didn't happen).

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Recall the inclusion-exclusion formula (with two sets):

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Additional properties

As a consequence of the general addition rule, we also have

(e) (Subadditivity.) $P(A \cup B) \leq P(A) + P(B)$.

(f) (Additivity.) If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

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$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{4} + \frac{3}{13} - \frac{3}{52} \\ &= \frac{11}{26} \approx 0.42 \end{aligned}$$

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Final answer. The probability of drawing either a heart or a face card is $\frac{11}{26}$, which is **less than** the sum of the individual probabilities because we subtract the overlap.

Group exercises

Random Group Assignments

Ashley, Isabelle: 3
Bailey, Tanna: 7
Blaisdell, Olyver: 11
Brown, Ashton: 5
Cardenas, Dominic: 6
Cutler, George: 9
Fabik, Roland: 10
Fellows, Audrey: 6
Fitzgerald, Ryan: 8
Fuller, Ryan: 2
Griffen, Clara: 7
Grutsch, Tanner: 6
Hager, Nathan: 10
Horner, Jayden: 11
Johnson, Jaret: 1
Kintzel, Samantha: 10
Lameres, Alexis: 2
Lee, Holden: 1
Mayala, Reece: 3
McLean, Connor: 3
Mcbride, Daniel: 9
Miller, Maverick: 9
Moss, Emily: 12
Mousseau, Matthew: 12
Neuman, Evan: 4
Putnam, John: 1
Ramos, Vydel: 5
Rathbun, Jedidiah: 4
Roberts, Matthew: 8
Roller, Nathan: 12
Ryan, Otis: 8
Shevtsov, Timothy: 11
Stensland, Emma: 7
Thiessen, Logan: 4
Thompson, Jaiden: 2
Vath, Ethan: 5

Group exercises

1. (More poker hands.) Give the probability of obtaining the various hands below in a game of 5-card poker.
    a. A **three of a kind**. This means three cards of the same rank and two other cards of different ranks, such as three 10s, a 7, and a jack.
    b. A **full house.** This is three cards with one common rank and two other cards of another common rank, such as three queens and two 4s.
    c. A **single pair.** This is two cards with the same rank and three other cards with three other ranks, such as two 9s, a king, an 8, and a 5.
    d. **Two pairs.** This is two cards with one common rank, two more cards with another common rank, and the remaining card has yet another rank, such as two jacks, two 8s, and a 3.
    e. A **flush.** This means all five cards have the same suit.
2. Prove Scheinerman Proposition 31.8. You may assume the the general addition rule (Scheinerman Proposition 31.7) holds.
3. Three dice are rolled. What is the probability that... (a) ... none of the dice shows 1? (b)... at least one die shows 1? (c) ... at least one die shows 2? (d) ... none of the dice shows 1 or 2? (e) ... at least one die shows 1 or at least one die shows 2 (or both)? (f) .... at least one die shows 1 and at least one die shows 2?

Solution to Group Exercise #1a

Problem. What is the probability of getting a **three of a kind** in a 5-card poker hand?

Solution. Let S be the sample space. That is,

$$S = \Big\{ \{A\spadesuit 2\spadesuit 3\spadesuit 4\spadesuit 5\spadesuit\},\ \ldots,\ \{9\heartsuit 10\heartsuit J\heartsuit Q\heartsuit K\heartsuit\} \Big\}.$$

Let $E \subset S$ be the event that the poker hand is a three of a kind. Then by the equally likely probability formula,

$$P(E) = \frac{|E|}{|S|} = \frac{\#\text{ of three of a kind poker hands}}{\#\text{ of poker hands}}$$

Now $|S| = \binom{52}{5} = 2,598,960$, so we just need to find $|E|$. We proceed in steps:

1. **Choose the rank** for the three matching cards (e.g., three Kings or three 7s). There are 13 possible ranks.
2. **Select 3 of the 4 suits** for that rank: $\binom{4}{3} = 4$.
3. **Choose two kicker cards** (the remaining two cards that don't match the three of a kind rank).
   a. **Choose 2 different ranks** from the remaining 12 ranks: $\binom{12}{2} = 66$
   b. **Select 1 suit** for each chosen rank: $\binom{4}{1} \times \binom{4}{1} = 16$
4. **Combine the choices**: $|E| = 13 \times 4 \times 66 \times 16 = 54,912$.

Final Answer. The probability of getting a three of a kind in a 5-card poker hand is $\frac{54,912}{2,598,960} \approx 0.021128$, or about 1 in 47 hands.

Solution to Group Exercise #1b

Problem. What is the probability of getting a **full house** in a 5-card poker hand?

Solution. Let S be the sample space. That is,

$$S = \Big\{ \{A\spadesuit 2\spadesuit 3\spadesuit 4\spadesuit 5\spadesuit\},\ \ldots,\ \{9\heartsuit 10\heartsuit J\heartsuit Q\heartsuit K\heartsuit\} \Big\}.$$

Let $E \subset S$ be the event that the poker hand is a full house. Then by the equally likely probability formula,

$$P(E) = \frac{|E|}{|S|} = \frac{\#\text{ of full house poker hands}}{\#\text{ of poker hands}}$$

Now $|S| = \binom{52}{5} = 2,598,960$, so we just need to find $|E|$. We proceed in steps:

1. **Choose the rank** for the three matching cards (e.g., three Kings or three 7s). There are 13 possible ranks.
2. **Select 3 of the 4 suits** for that rank: $\binom{4}{3} = 4$.
3. **Choose the rank** for the two matching cards (e.g., two Queens or two 6s). There are 12 possible ranks which differ from the three-of-a-kind rank.
4. **Select 2 of the 4 suits** for that rank: $\binom{4}{2} = 6$.
5. **Combine the choices**: $|E| = 13 \times 4 \times 12 \times 6 = 3,744$.

Final Answer. The probability of getting a three of a kind in a 5-card poker hand is $\frac{3,744}{2,598,960} \approx 0.00144$, or about 1 in 694 hands.

Solution to Group Exercise #1c

Problem. What is the probability of getting a **single pair** in a 5-card poker hand?

Solution. Let S be the sample space. That is,

$$S = \Big\{ \{A\spadesuit 2\spadesuit 3\spadesuit 4\spadesuit 5\spadesuit\},\ \ldots,\ \{9\heartsuit 10\heartsuit J\heartsuit Q\heartsuit K\heartsuit\} \Big\}.$$

Let $E \subset S$ be the event that the poker hand is a single pair. Then by the equally likely probability formula,

$$P(E) = \frac{|E|}{|S|} = \frac{\#\text{ of single pair poker hands}}{\#\text{ of poker hands}}$$

Now $|S| = \binom{52}{5} = 2,598,960$, so we just need to find $|E|$. We proceed in steps:

1. **Choose the rank** for the two matching cards (e.g., two Kings or two 7s). There are 13 possible ranks.
2. **Select 2 of the 4 suits** for that rank: $\binom{4}{2} = 6$.
3. **Select 3 other ranks** for the remaining 3 cards (out of the 12 ranks which differ from the single pair rank): $\binom{12}{3} = 220$.
4. **Select 1 suit for each of the 3 different ranks**: $\binom{4}{1} \times \binom{4}{1} \times \binom{4}{1} = 64$.
5. **Combine the choices**: $|E| = 13 \times 6 \times 220 \times 64 = 1,098,240$.

Final Answer. The probability of getting a single pair in a 5-card poker hand is $\frac{1,098,240}{2,598,960} \approx 0.42257$, or about 1 in 2.37 hands.

Solution to Group Exercise #1d

Problem. What is the probability of getting **two pairs** in a 5-card poker hand?

Solution. Let S be the sample space of poker hands. Let $E \subset S$ be the event that the poker hand is two pairs. Then by the equally likely probability formula,

$$P(E) = \frac{|E|}{|S|} = \frac{\#\text{ of two pair poker hands}}{\#\text{ of poker hands}}$$

Now $|S| = \binom{52}{5} = 2,598,960$, so we just need to find $|E|$. We proceed in steps:

1. **Choose ranks** for the pairs (e.g., we might have a pair of Kings and pair of 7s): $\binom{13}{2} = 78$.
2. **Select 2 of the 4 suits** for each of those ranks: $\binom{4}{2}\binom{4}{2} = 6 \times 6$.
3. **Choose the rank** for the fifth card. There are $\binom{11}{1} = 11$ ranks which differ from those of the two pairs.
4. **Select 1 of the 4 suits** for that rank: $\binom{4}{1} = 4$.
5. **Combine the choices**: $|E| = 78 \times 6 \times 6 \times 11 \times 4 = 123,552$.

Final Answer. The probability of getting two pairs in a 5-card poker hand is $\frac{123,552}{2,598,960} \approx 0.0475$, or about 1 in 21 hands.

Solution to Group Exercise #1e

Problem. What is the probability of getting a **flush** in a 5-card poker hand?

Remark. A **flush** consists of five cards of the *same suit*, but *not in sequential order*. For example: 2♠5♠8♠J♠Q♠. If the cards are in sequence, it would be a **straight flush** instead.

Solution. Let S be the sample space of poker hands. Let $E \subset S$ be the event that the poker hand is a flush. We write $E = A - B$, where A is the number of flushes (including straight flushes), and B is the number of straight flushes. Hence, $|E| = |A| - |B|$.

To find $|A|$:

1. **Choose the suit**: There are 4 suits.
2. **Select 5 cards from the suit**: $\binom{13}{5} = 1,287$.
3. **Combine the choices**: $|A| = 4 \times 1287 = 5,148$.

To find $|B|$:

1. **Choose the suit**: There are 4 suits.
2. **Select the straight flush in the suit**: For each suit, there are 10 straight flushes, (from $A-2-3-4-5$ to $10-J-Q-K-A$).
3. **Combine the choices**: $|B| = 4 \times 10 = 40$.

Hence, the number of valid flush hands is $|E| = |A| - |B| = 5,148 - 40 = 5,108$.

Final Answer. The probability of getting a flush is $\frac{5,108}{2,598,960} \approx 0.001965$, or about 1 in 509 hands.

Solution to Group Exercise #2

Problem. A Let (S, P) be a probability space. Then

(a) (Probability of entire space.) $P(S) = 1$.
(b) (Probability of empty set.) $P(\emptyset) = 0$.
(c) (Probability of complement.) $P(\overline{A}) = 1 - P(A)$,
where $\overline{A} := S - A$ is called the *complement* of A (i.e. A didn't happen).
(d) (General addition rule.) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$,
where A and B be events.
(e) (Subadditivity.) $P(A \cup B) \leq P(A) + P(B)$.
(f) (Additivity.) If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

Assume (d) above and prove the rest.

Solution.

(a) By the defs. of probability space and event, $P(S) = \sum_{s \in S} P(s) = 1$.
(b) By the defs. of probability space and event, $P(\emptyset) = 0 = \sum_{s \in \emptyset} P(s) = 0$.
(c) Using the general addition rule with $B = \overline{A}$:

$$\begin{aligned} & P(A \cup \overline{A}) = P(A) + P(\overline{A}) - P(A \cap \overline{A}) \\ \implies & P(S) = P(A) + P(\overline{A}) - P(\emptyset) \\ \implies & 1 = P(A) + P(\overline{A}) - 0 \qquad \text{by (a), (b)} \end{aligned}$$

Algebra gives the result.
(e) This holds by the general addition rule, noting that $P(A \cap B) \geq 0$.
(f) If $A \cap B = \emptyset$, then $P(A \cap B) = 0$ by part (b). Hence, the result follows by the general addition rule.

Solution to Group Exercise #3

Problem. Three dice are rolled. What is the probability that...

a.) ... none of the dice shows 1?

b.) ... at least one die shows 1?

c.) ... at least one die shows 2?

d.) ... none of the dice shows 1 or 2?

e.) ... at least one die shows 1 or at least one die shows 2 (or both)?

f.) .... at least one die shows 1 and at least one die shows 2?

Solution.

Define the following events $A =$ at least one die shows 1 $B =$ at least one die shows 2

So the complementary events are $\overline{A} =$ none of the dice show 1 $\overline{B} =$ none of the dice show 2

a.) The outcomes in event $\overline{A}$ are the lists of length 3 with outcomes chosen from the set $\{2,3,4,5,6\}$. Hence $|A| = 5^3$. By the equally likely probability formula, $P(\overline{A}) = \frac{|A|}{|S|} = \frac{5^3}{6^3} \approx 0.5787$.

b.) By the probability of a complementary event, $P(A) = 1 - P(\overline{A}) \approx 1 - 0.5787 = 0.4213$.

c.) $P(B) = P(A) \approx 0.4213$.

d.) The outcomes in event $\overline{A} \cap \overline{B}$ are the lists of length 3 with outcomes chosen from the set $\{3,4,5,6\}$. Hence $|\overline{A} \cap \overline{B}| = 4^3$. By the equally likely probability formula, $P(\overline{A} \cap \overline{B}) = \frac{|\overline{A} \cap \overline{B}|}{|S|} = \frac{4^3}{6^3} \approx 0.2962$.

e.) By the probability of a complementary event, $P(A \cup B) = 1 - P(\overline{A \cup B})$. By DeMorgan's laws for sets (or intuition), $\overline{A \cup B} = \overline{A} \cap \overline{B}$. Hence, using part (d), $P(\overline{A \cup B}) = 0.2962$, and so $P(A \cup B) = 1 - P(\overline{A \cup B}) \approx 0.7038$.

f.) By the general addition law, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$. Filling in results from (b), (c), and (e), we find $P(A \cap B) \approx 0.4213 + 0.4213 - 0.7038 = 0.1388$.